

Project 16

Orientation-Dependent Connectivity on custom_6R

Format: Group of two students; simulation-based project.

Course foundations: Lectures 4–7; Exercises 2–3.

Motivation

The visible spatial trace of a tool does not fully describe its task. For the same wrist-center curve, different orientation schedules can create different wrist singularities and feasible joint-space continuations. This project studies that dependence while holding the positioning problem fixed.

Task

Overall objectives. Compare a small set of orientation schedules on one custom_6R wrist-center curve and explain how they alter the feasible continuations of its arm and wrist IK solutions.

- **Validate the robot and the chosen curve.** Use the exact custom_6R model, with the custom_3R positioning arm and spherical wrist. Check the D–H/URDF correspondence and fixed tool transform. Choose one short closed wrist-center curve from a supplied feasible-path example.
- **Map the common arm problem.** Plot the positioning determinant and zero contours in (q_2, q_3) and map the two-/four-arm-IK workspace regions. Enumerate the initial arm IKs and continue the arm candidates over the curve. Overlay these trajectories so that all orientation experiments share one clearly documented positioning baseline.
- **Define three orientation schedules.** Compare a constant tool orientation, that orientation composed with one fixed offset rotation, and a smooth periodic orientation variation. For example use $R(s) = R_0 \exp(\beta \sin(2\pi s)\hat{u})$ with fixed unit axis u . Record the angle and derive tool position consistently from the wrist center.
- **Lift and classify the full-pose candidates.** Complete each continued arm candidate with the wrist solutions and track each starting flip separately. Compare full-pose residuals, wrist/arm regularity, joint limits, and endpoint configuration. Do not interpret an orientation change as permission to jump between unrelated IKs.
- **Explain changes in connectivity.** Build a compact chart of path progress versus candidate identity for each schedule, marking feasible segments and failure reasons. Compare where a complete arm continuation loses a full-pose lift. Refine one boundary and explain it using the corresponding wrist singularity factor or active joint limit.

Specifics and scope

Use one wrist-center curve, three orientation schedules, and all recovered regular initial candidates (up to eight). Begin with approximately 150 samples. The course arm/wrist IK and a reference curve are available for reuse. Investigate the prescribed schedules; optimizing over arbitrary orientation functions is outside scope.

Check both translation and orientation residuals for every claimed full-pose IK. Document angle periodicity and limits, Jacobian conventions, and any scaling used for singular values. State all fixed coordinates in reduced plots; a two-dimensional slice does not show the entire 6R singular set. Refine decisive transitions, and distinguish sampled evidence from a continuous-path guarantee. Collision freedom is not implied by these kinematic checks.

Expected outcome

Explain why the same positioning task can have different full-pose feasibility depending on orientation. Deliver the common arm maps, three explicit pose schedules, candidate feasibility charts, and a comparison of the global joint-space continuations.

Submit reproducible code, model parameters and test data, the required plots, a concise 5–6-page report, and a short simulation demonstration. Explain both successful cases and failures; conclusions must follow the numerical and geometric evidence rather than an expected outcome.

Resources

The `custom_6R` consists of the `custom_3R` positioning arm plus a spherical wrist. The arm is orthogonal with non-intersecting consecutive axes. Its URDF is `lectures_main/assets/models/custom_6R/custom_6R_new.urdf`; the corresponding STL meshes are in that directory. Inspect wrist-axis concurrency and base/tool transforms using the lecture website. Use Lectures 4–6 for the custom arm and wrist structure, and Lecture 7 for continuation. The supplied reference curve must be interpreted in the exact `custom_6R` wrist-center frame. Existing position-only visualizations must be complemented with full-pose validation.

Use the lecture website to verify D–H parameters, visualize IKs and singularities, and simulate paths. All listed paths are relative to the repository root. Start the website following `lectures_main/README.md`.

Workload and collaboration

Budget approximately 60 hours per student (120 person-hours per pair) for this 2-ECTS project, following EPFL’s 30-hours-per-credit convention.¹ Deduct any lectures or exercises included in those credits. Allocate roughly 20 team-hours to model validation, 50 to the core work, 30 to experiments, and 20 to reporting. Reuse the specified IK and simulations. Share validation and interpretation even if modelling and implementation are divided between students. Hardware execution is outside scope.

¹EPFL: Design a Course, workload guidance.